3(a)

momentum after elastic collision	= - mu	[1]
time between collisions	= 2L/u	[1]
number of collisions per unit time	= u/2L	[1]
rate of change of momentum	= - mu^2/L	[1]
average force	= mu^2/L	[1]

(b)(i) Gas molecules collide with one another elastically. [1]

(ii) Since collisions are elastic, the average speed of gas molecules incident on the wall remains the same. [1]

So frequency of collision between each wall and the molecule also remains unchanged. [1]

Therefore the pressure exerted on the wall is not affected.

(c) Given in (b) $pV = 1/3\ Nm\langle c^2\rangle$ - - - - (1)

By Ideal Gas Law: $pV = NkT$ - - - - (2) [1]

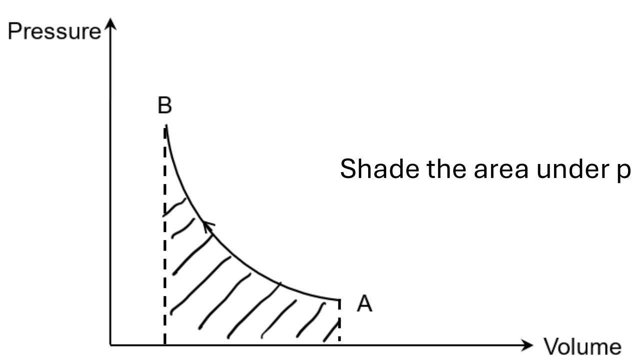
Equating (1) = (2) $1/3\ Nm\langle c^2\rangle = NkT$

=> $½\ m\langle c^2\rangle = 3/2\ kT$ [1]

Hence average ke, $½\ m\langle c^2\rangle = 3/2\ kT$

$\therefore$ Ave KE $\propto$ T (because k is a constant)

(d)(i)1.



Shade the area under p-V graph [1]

Fig 3.2

2. Since the process A -> B is such that $\Delta U = 0$ (isothermal), and $U = \frac{3pV}{2}$ then

$p_A V_A = p_B V_B$. [1]

Hence there is no difference in the product $p_A V_A$ and $p_B V_B$. [1]

(ii) If the change takes place very quickly, there is no time for heat transfer with the surrounding. So it is an adiabatic change. i.e. $Q = 0$. [1]

Fig. 3.2 shows a compression. i.e. W is positive. So $\Delta U = Q + W$ is also positive. i.e. U increases. [1]

Since $U \propto c_{rms}^2$ of the molecules, it means the mean square speed of the molecules increases. [1]

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4(a) Since the tube is in equilibrium